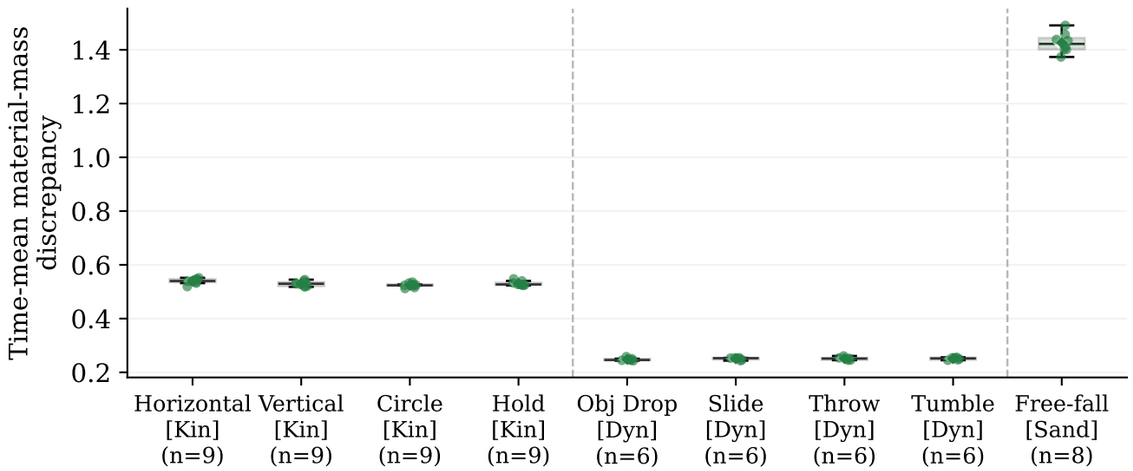
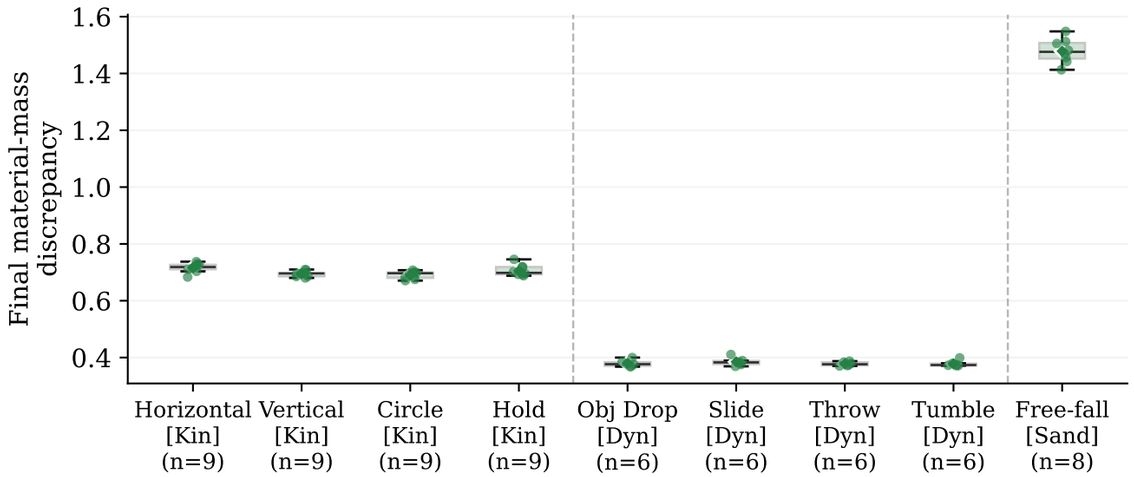
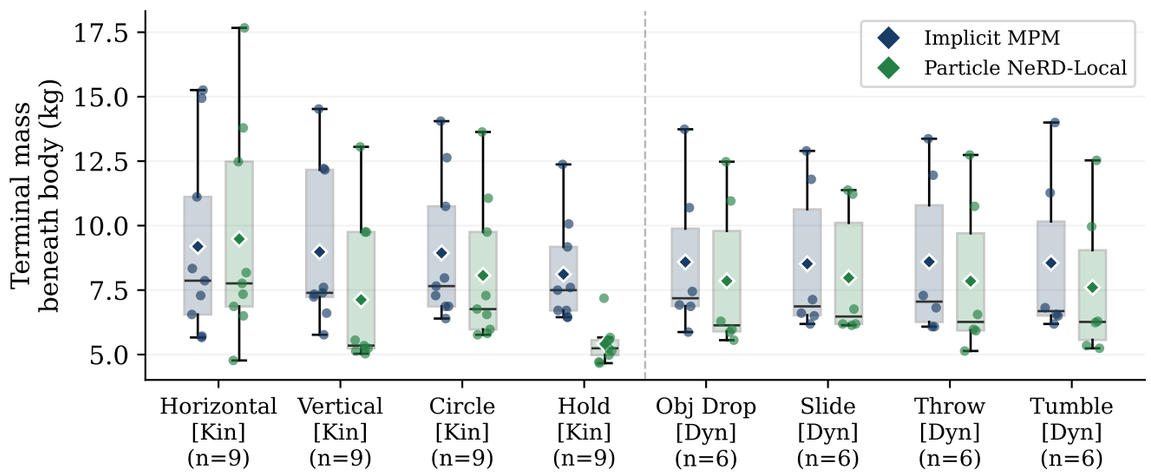
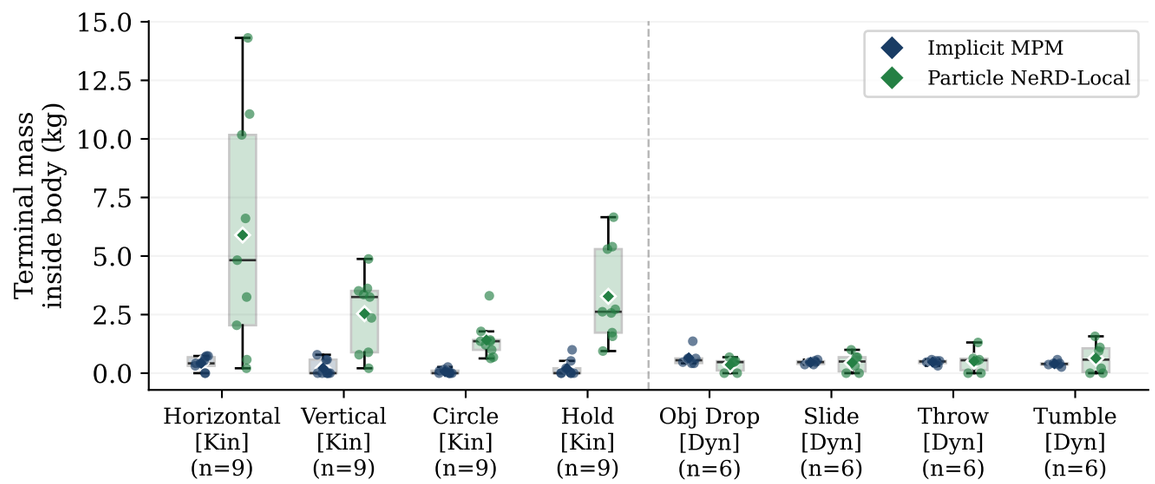
By Motion Type — Kinematic Object Motions (Horizontal, Vertical, Circle, Hold) | Dynamic Object Motions (Obj Drop, Slide, Throw, Tumble) | Sand Free-Fall (Flat Ground)









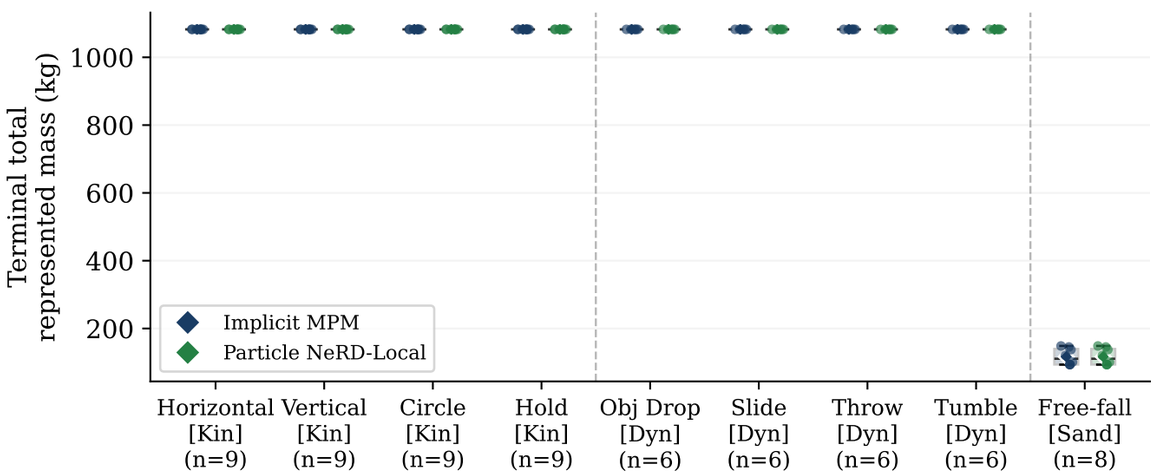


Figure 3b: Frozen object-model checkpoint 10,800. Material-mass discrepancy and represented mass grouped by motion: Kinematic object motions (Horizontal, Vertical, Circle, Hold), Dynamic object motions (Obj Drop, Slide, Throw, Tumble), and Unconfined Sand Free-Fall on flat ground.